

Note on First Principles in Arithmetic Expression Geometry: F_2 , Canonical Forms, Condensation, and the Emergence of Space

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Abstract

This note summarizes a discussion on foundational, "first principles" thinking for Arithmetic Expression Geometry (AEG). The core idea posits the free group on two generators, F_2 , as the origin of computational complexity. It proposes that the introduction of canonical forms for arithmetic expressions acts as a "seed for spatiality," with these forms "weaving" a spatial grid, defining a "geometrization condition." Computation is then conceptualized as flow on this grid, and computational complexity as the length of this flow. A significant extension to this first principle is the idea that canonical forms, viewed as "points" in this space, encapsulate "computational energy," thereby possessing "computational mass" and inducing "spatial curvature," drawing an analogy to general relativity. The concept of "Condensation" is central, understood as a conceptual-level operation transforming a "pure time series" of basic actions into a "spacetime node," enabling the interpretation of the remaining operational network as a geometric grid. Algebraically, this is often realized by quotienting F_2 by the normal closure of a set of relators (canonical forms), which act as "condensation nuclei." More broadly, any equivalence relation identifying "operational history" with "position" and introducing "spatial dimensions" is a form of condensation. Thus, condensation is fundamentally a metaphor for "space emerging from time," with group quotients being its most concise algebraic projection. The Baumslag-Solitar group $BS(1, 1)$ (the free abelian group $\mathbb{Z}^2$) is analyzed as the simplest case, where the condensed canonical form is the commutator. The global arithmetic torsion of the commutator $[x, y]$ (where $x = \oplus_\mu, y = \otimes_\lambda$) is algebraically calculated as $\tau_{\text{alg}}([x, y]) = \mu(e^{-\lambda} - e^\lambda)$.

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1 Introduction: Seeking First Principles

This note captures the essence of a discussion aimed at unearthing "first principles" to underpin the framework of Arithmetic Expression Geometry (AEG). The driving motivation is to move beyond constructing various AEG spaces ad hoc and instead to understand how such geometric structures might naturally emerge from fundamental algebraic and computational concepts. The discussion explored the role of the free group F_2 , the significance of canonical forms, the concept of "condensation" in forming specific group structures, and a novel analogy to general relativity where computational entities might induce geometric curvature.

2 Core Tenets of the First Principle

The proposed first principle thinking can be summarized through several interconnected ideas:

1. **F_2 as the Primordial Source of Complexity:** The free group on two generators, $F_2 = \langle \mathbf{x}, \mathbf{y} \rangle$, is posited as the fundamental origin of both time complexity (sequence length) and space complexity (structural arrangement) in arithmetic processes. $\mathbf{x}$ and $\mathbf{y}$ represent abstract elementary operations (e.g., a unit addition and a unit multiplication).
2. **Canonical Forms as Seeds of Spatiality and the Geometrization Condition:** The act of choosing or defining a *canonical form* for arithmetic expressions is a foundational step that "implants a seed of spatiality." The collection of these canonical forms, and the way they relate to each other (how they "join hands"), is what *weaves the fabric of a geometric space*. This process itself can be understood as a "geometrization condition."
3. **Computation as Flow on a Geometrized Grid:** Once this geometric space (or grid) is established by the interlinking of canonical forms, any specific computation can be visualized as a *flow* or a path on this grid.
4. **Complexity as Path Length:** The *length* of this flow path, measured according to a metric inherent to the geometrization, corresponds to the computational complexity.
5. **Canonical Forms as the Basis for Spacetime Transformation in Computation:** The chosen canonical forms are fundamental in establishing the relationship and potential transformations between the temporal aspect of computation and its spatial realization.

3 Extension: Canonical Forms, Computational Energy, and Spatial Curvature

A significant extension to the above first principles emerged:

1. **Canonical Forms as Points with "Computational Energy"**: Each canonical form, viewed as a "point" in the emergent geometric space, encapsulates "computational energy."
2. **"Computational Mass" from Energy**: This computational energy endows the canonical form with a "computational mass."
3. **Spatial Curvature Induced by Computational Mass**: Analogous to General Relativity, the "computational mass" of these canonical-form-points dictates how the arithmetic expression space itself is curved.
4. **Space Complexity and Curvature**: Space complexity may be intrinsically linked to this induced curvature.

This perspective shifts AEG from describing arithmetic processes within a pre-defined geometry to a framework where geometry itself is an emergent property, shaped by the "matter" (canonical forms with computational energy) of arithmetic.

4 Condensation: The Emergence of Space from Time

The concept of "condensation" is pivotal in understanding how structured geometric spaces emerge from the purely sequential (temporal) nature of operations in F_2 .

Definition 4.1 (Condensation in AEG Context). In the AEG context, *condensation* is a **conceptual-level** operation: it "condenses" a *pure time series* (a sequence of elementary actions) into a *spacetime node*, enabling the remaining network of operations to be interpreted as a geometric grid.

- **Algebraic Realization.** The most economical technical realization is to quotient the free group F_2 by the normal closure N of a set of relators, yielding $G \simeq F_2/N$. These canonical forms that are folded into the identity element are precisely the "condensation nuclei."
- **Conceptual Extension.** Any equivalence relation that identifies "operational history" with "position" be it congruence in rewriting systems, β -reduction in programming semantics, or homotopy collapse in topology can be considered a form of condensation, as long as it introduces an additional "spatial dimension" to the operations.

Therefore, *condensation is first and foremost a metaphor for "space emerging from time"*; group quotients are merely its most concise algebraic projection.

This definition (Definition 4.1) emphasizes that imposing relations on F_2 (the algebraic act of forming a quotient group $G = F_2/N$) is a concrete way to achieve condensation. The canonical forms of the elements in N (the relators) are the "pure time series" that are conceptually "condensed" into a single point (the identity element in G). This act of condensation, by identifying previously distinct operational sequences, forces the remaining distinguishable sequences to arrange themselves in a way that reveals a "spatial" structure (the geometric grid of G).

4.1 The Case of $BS(1, 1) \cong \mathbb{Z}^2$

$BS(1, 1)$ is the free abelian group on two generators, $\mathbb{Z}^2$. It is obtained from $F_2 = \langle \mathbf{x}, \mathbf{y} \rangle$ by adding the commutation relation $[\mathbf{x}, \mathbf{y}] = \mathbf{xyx}^{-1}\mathbf{y}^{-1} = 1$.

- **Condensed Canonical Form (Condensation Nucleus):** The commutator $[\mathbf{x}, \mathbf{y}]$.
- **Algebraic Effect:** Operations corresponding to $\mathbf{x}$ and $\mathbf{y}$ become commutative in $BS(1, 1)$. The normal subgroup N is the commutator subgroup F_2' .
- **Geometric Effect:** The Cayley graph of F_2 (an infinite 4-valent tree) "collapses" or "folds" (condenses) into the standard square grid of $\mathbb{Z}^2$. All loops defined by the commutator relation become contractible (condensed to a point).
- **AEG Interpretation:**
 - The Accumulative Commutative Space (ACS) directly corresponds to this $\mathbb{Z}^2$ grid.
 - The global arithmetic torsion for any path in a geometry faithfully representing $BS(1, 1)$ should effectively be zero.
 - An AEG Expression Space ($\mathfrak{E}$) realizing $BS(1, 1)$ would be flat.

5 Global Arithmetic Torsion of the Commutator Revisited

The global arithmetic torsion for the commutator path $\gamma_c = [\mathbf{x}, \mathbf{y}] = \mathbf{xyx}^{-1}\mathbf{y}^{-1}$ (where $\mathbf{x} = \oplus_\mu, \mathbf{y} = \otimes_\lambda$) was algebraically calculated as:

$$\tau_{\text{alg}}([\mathbf{x}, \mathbf{y}]) = \mu(e^{-\lambda} - e^\lambda)$$

This is based on the definition $\tau_{\text{alg}}(\gamma) = \nu(\gamma)(v_0) - \nu(\bar{\gamma})(v_0)$, where $\bar{\gamma}$ is the path with its literal sequence of operations applied in reverse order. This result is distinct from the locally defined arithmetic torsion $\tau_{\text{local}} = \mu(e^\lambda - 1)$, which compares $\nu(\mathbf{yx})(v_0)$ with $\nu(\mathbf{xy})(v_0)$. The Triple Identity in AEG must relate $\tau_{\text{alg}}(\gamma)$ to the geometric integral forms in the ACS, requiring careful definition of the integration domain Σ_γ based on the ACS traces of both γ and $\bar{\gamma}$.

6 Concluding Remarks and Path Forward

The "first principles" framework, enriched by the conceptual definition of "condensation," offers a powerful lens through which to view the emergence of geometric structures in AEG. The idea that canonical forms (especially those corresponding to relations) act as "condensation nuclei," possessing "computational energy/mass" and shaping spatial curvature, remains a highly stimulating, albeit currently metaphorical, avenue for future research.

The immediate challenge is to make these metaphors concrete. The analysis of $BS(1, 1)$ serves as a foundational step, illustrating how the condensation of the commutator (a "pure time series" of operations) leads to a simple spatial grid ($\mathbb{Z}^2$). The

discrepancy observed in torsion calculations underscores the need for rigorous and consistent application of definitions when bridging algebraic and geometric manifestations.

To "tear through the superficial layer and enter a new conceptual space," the path forward might involve:

1. **Formalizing "Computational Energy/Mass":** Develop a concrete, quantifiable definition for the "energy" or "mass" of a canonical form (perhaps related to its ACS coordinates, its F_2 word length, or the complexity of the sub-network it condenses).
2. **Mechanism of Curvature Induction:** Explore how these "masses" might define a metric or connection on the space of canonical forms, leading to a calculable notion of curvature. Could tools from discrete differential geometry or statistical mechanics of networks be relevant?
3. **Revisiting $\mathfrak{E}_0$ and $\mathfrak{E}_1$ through Condensation:**
 - For $\mathfrak{E}_0$ (Poincaré disk, single zero point): What "time series" is condensed to form the central point singularity? How does the "fire-burning" propagation of the assignment function relate to the idea of flow on a grid woven by other, non-singular canonical forms? Does the radial symmetry of $\mathfrak{E}_0$ imply a highly symmetric condensation process?
 - For $\mathfrak{E}_1$ (upper half-plane, realizing $BS(2, 1)$ or $BS(1, 2)$): We know the condensed relation is $b^{-1}a^mba^{-n} = 1$. How does this specific "condensation nucleus" lead to the particular hyperbolic geometry and the rectilinear/transformed grids observed? Can the constant negative curvature of $\mathfrak{E}_1$ be understood as arising from a specific, uniform distribution or interaction of these condensed "energies"?
4. **The Role of the Encoding Space:** Re-evaluate the necessity and structure of the "Encoding Space" proposed earlier. Perhaps the "condensation" metaphor itself, when fully developed, provides the necessary bridge between the raw operations of F_2 and the structured (and possibly curved) Expression Space, with ACS remaining the final commutative projection. The process of condensation might be precisely what the Encoding Space was trying to capture the rules by which "time" (sequences) gets mapped to "space" (positions/nodes).

The conceptual leap lies in seeing the "relations" not just as constraints, but as active agents (the "condensation nuclei") that *create* the geometric landscape in which arithmetic processes unfold.